Rodeo NOAA mission with SeaExplorer glider

Technical report

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Diffusion	Interne	ALSEAMAR
	Externe	NOAA

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INTRODUCTION

The present document is a technical report related to the preparation and the conduct of the glider mission in Hawaii from January 28,2026 to February 11,2026.

The mission aims to advance the monitoring of marine mammals in US waters. To this end, eight gliders of different brands and generations were deployed over a two-week period and were required to follow the same mission plan. All eight gliders were equipped with PAM (passive acoustic monitoring).

The SeaExplorer SEA117 glider used for the mission had an algorithm for real-time detection and classification of marine mammals and monitoring of ambient noise.

This is technical report aims to address:

- Technical information about the SeaExplorer glider and sensors used
- Technical information about the glider mission
- Work conducted in support of the mission operation (preparation, logistic)

1 SEAEXPLORER GLIDER TECHNOLOGY

The main working principle of the SeaExplorer glider are briefly presented in this section. For further details, please refer to Appendix 1. The SeaExplorer underwater glider (Fig. 1) is designed and manufactured by ALSEAMAR. It is an autonomous platform driven by buoyancy changes (*i.e.* no propeller). Its endurance ranges from weeks to months, depending on the mission design, the payload and the battery pack. The glider is thus periodically recharged (Li-Ion batteries), this operation lasts less than 24 hours that allow to complete an uninterrupted time-series.

Figure 1: SeaExplorer glider

The classical navigation mode of the SeaExplorer glider is a saw-tooth like trajectory (Fig. 2 right) but various dive profiles (yo) are possibles :

- Simple yo : 1 ascent / 1 descent in a defined layer
- Multi-yos : several ascent/descent before surfacing in a defined layer
- Drift : glider drift with the current at a fixed depth
- Bottom landing (the glider land at the seafloor for a fixed duration)

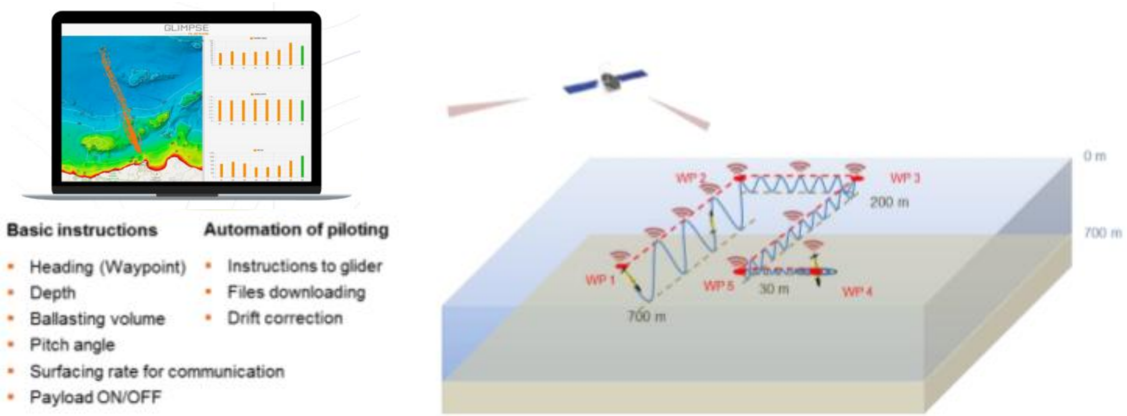

Figure 2: Communicating with a SeaExplorer

Along its route, the glider collects environmental data at high spatial (order meter) and temporal (order second) resolution, within the water-column (maximum depth is 1000m). Various physical, biogeochemical and acoustic miniaturized sensors are commercially available and integrated on the SeaExplorer (see Appendix 1 for the list of sensors). Prior to a given mission, the appropriate suite of sensors is selected to meet the scientific objectives of the project.

At each surfacing, the glider is monitored by the onshore based ALSEAMAR pilot team (24/24hours 7/7days through the web interface GLIMPSE, Fig. 4). This is done through double-way IRIDIUM telecommunication. This allows to check for the main operating parameters (glider position, autonomy, behavior), scientific data and also adjust the mission design in near real time.

For this project, all NOAA partner received a piloting access of the GLIMPSE interface for comparison with other technologies.

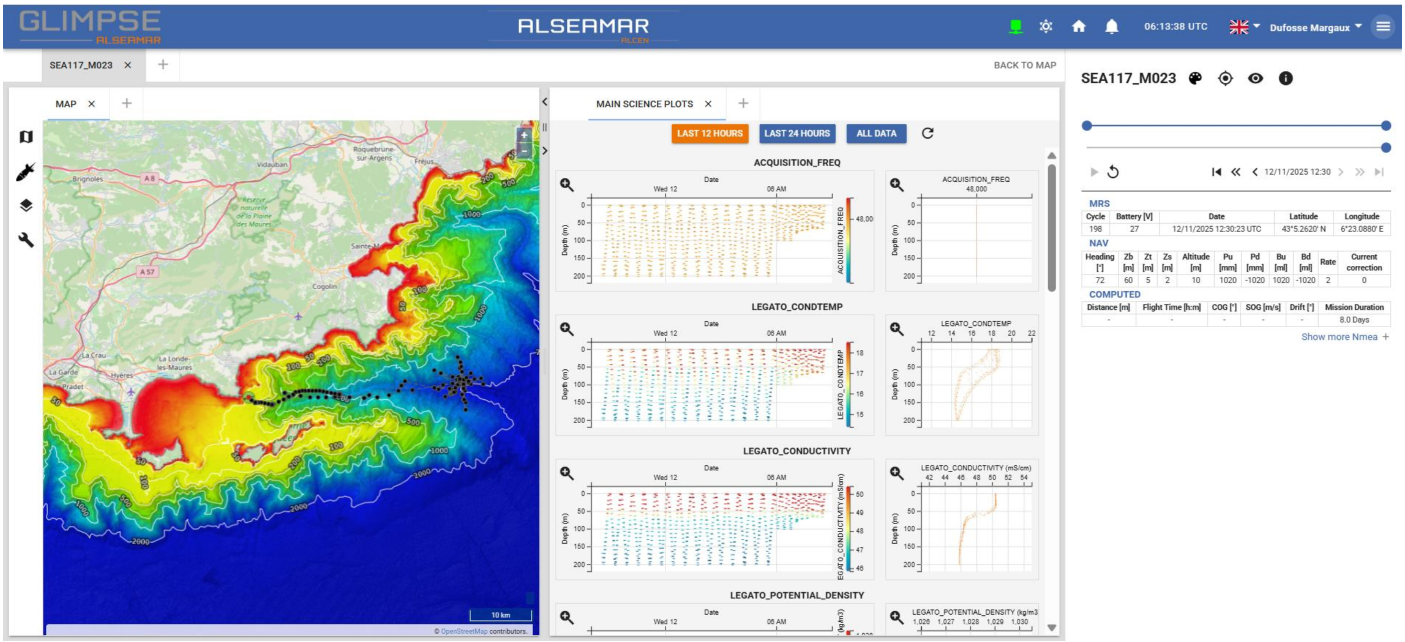

Figure 4: Glimpse piloting platform. Trajectory is from the SEA117 which correspond to the Sea Acceptance Test mission in the Mediterranean Sea realized before the campaign.

2 PAYLOAD DEDICATED TO THE MISSION

One SeaExplorer underwater glider (SEA117) was mobilized for the purpose of the present project. The suit of sensors was selected in agreement with the NOAA (Table 1).

All sensors were mechanically integrated to the glider, cleaned and protected following best practices and manufacturer's recommendations. Sensors are factory calibrated and configured for the mission. A short description of selected sensors is given below and further technical information (datasheet, calibration sheet) can be found in Appendix 2.

Table 1: Glider SEA117 sensors configuration

Sensor	SN	Manufacturer	Measured parameters
LEGATO CTD	230630	RBR	Conductivity, temperature and pressure
AURIS ACOUSTIC ANTENNA	666	ALSEAMAR	Acoustic antenna with 4 hydrophones

3 SENSOR DESCRIPTION

There are more information and calibration sheet about this differents sensor in *Appendix 2.*

3.1 LEGATO

SEA117 was equipped with a LEGATO RBR sensor. The data measured by the RBR LEGATO are pressure, temperature, conductivity and salinity. The sensor is integrated on the puck. Total power consumption is 190 mW. Density and celerity are derived from this data, and are now calculated and integrated into the GLIMPSE control platform in real time.

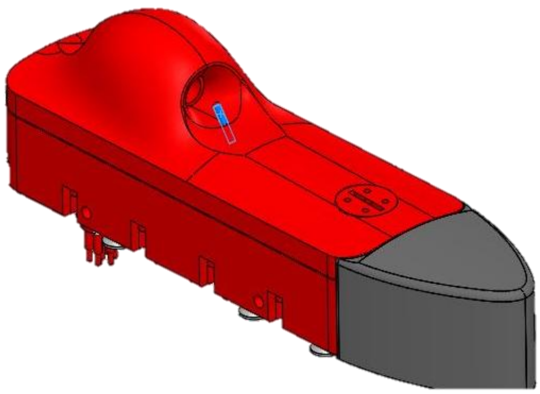

Figure 5 : LEGATO sensor

3.2 AURIS ACOUSTIC ANTENNA

SEA117 was equipped with an acoustic antenna designed by ALSEAMAR. The version chosen for this mission is the antenna made up of 4 hydrophones in a tetrahedral configuration. The AURIS on-board processing platform can also be used. The data is temporarily stored on an SD card at each yo and transferred to a storage hard disk at each surfacing. There are more information in *Appendix 2.*

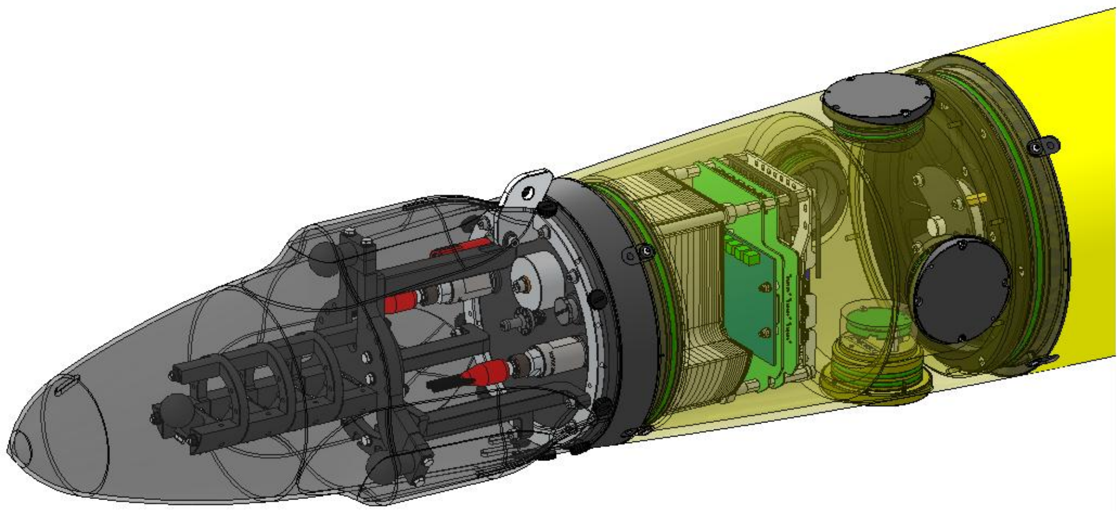

Figure 6 : AURIS sensor

4 PRE-MISSION PREPARATION

4.1 Pool ballasting

Before deployment, the glider was ballasted in the ALSEAMAR pool (*Fig 7*), according to the expected *in-situ* density that would be found in the survey area. This was done using publicly available database, the glider was ballasted to be neutral at 1023 kg.m^{-3}. The ballasting sheet can be found in Appendix 3.

Figure 7: Glider during ballasting

4.2 Magnetic calibration of the compass

For navigation purpose, gliders are equipped with compass. Raw heading data are generally biased because of sensor drift and magnetic disturbance induced by the glider itself (hard iron effect). Thus, it is required to calibrated compass before each mission. This was done at ALSEAMAR facilities before the SAT of gliders, outside and far from any magnetic disturbance. The procedure includes various rotation of the glider around the z-axis (heading) and the x-axis (roll). At the end of the procedure, calibrated data are validated by comparing with a reference hand-compass at every 45° (tolerance difference ~3° in heading). The calibration sheet can be found in Appendix 5.

4.3 Mission simulation

Once the glider was prepared and in its final configuration, it is put on simulation mode. This means that the glider realizes a virtual mission to test actuators (e.g. battery movements), communication and data logging. The duration of this test last from hours to days.

4.4 At sea test in the Mediterranean Sea

Once simulated mission is validated, the gliders are deployed in the Mediterranean Sea for an at sea acceptance test. This test was conducted in November, 2025 off Cavalaire using ALSEAMAR boat (Fig. 8). All the modes of navigation (deep and shallow yos, single and multi-yos) and sensor's functioning were tested and validated. The mission preparation sheet can be found in Appendix 4.

Figure 8. TANGO 3, ALSEAMAR boat used to realize the test mission in Mediterranean Sea.

4.5 Logistic

Logistics transportation for mobilization and demobilization between ALSEAMAR in Rousset, France and NOAA lab in Honolulu, Hawaii is carried out by air.

An ALSEAMAR employee was on site for the deployment and recovery of the glider.

The glider was piloting from France by various ALSEAMAR pilots according to the mission plan developed by NOAA.

5 MISSIONS MONITORING

5.1 Mission design

The mission design was implemented by NOAA for compare the acoustic detection of 8 different gliders. A precise timetable was put in place with different navigation phases and timetables to be followed.

The different objectives were:

1) Integration of real-time detection and classification algorithms for marine mammals

2) Monitoring of ambient noise

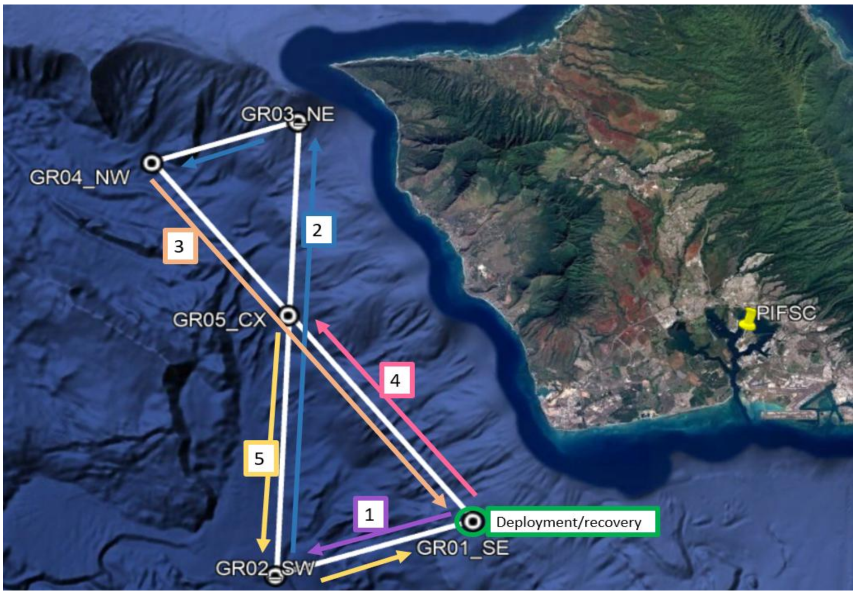

Figure 9: General overview of the mission plan

Table 3: Coordinates of mission plan waypoint

Name	Latitude_DDMM	Longitude_DDMM	Latitude_DD	Longitude_DD	Notes
GR01	21 13.6076	-158 10.2274	21.2267933	-158.1704567	SE; ~1190 m depth
GR02	21 11.0925	-158 19.3966	21.1848750	-158.3232767	SW; ~2990 m depth
GR03	21 31.9611	-158 18.3548	21.5326850	-158.3059133	NE; ~602 m depth
GR04	21 30.0704	-158 25.1493	21.5011733	-158.4191550	NW; ~2350 m depth
GR05	21 23.0629	-158 18.7956	21.3843817	-158.3132600	CX; ~2000 m depth
RECV	21 13.6076	-158 10.2274	21.2267933	-158.1704567	Same as GR01

5.2 Deployment

The January 28, 2026 the SEA117 SeaExplorer glider was deployed by ALSEAMAR employee and the team on the boat (*Fig 10*). On the start of the mission, the glider realized one shallow yos (30m) for equilibration purpose and check if all is good (navigation and scientific data). From yo 2, and after checking for navigation and scientific data, the mission planning was activated to realize the survey. Maximum depth was fixed to 1000 meters.

Figure 10: Picture of the deployment

5.3 Mission time-line of SEA117

Start date	End date	Duration (day)	Number of the mission	Number of cycles	Distance (km)
28/01/2026	11/02/2026	14	26	159	325.09

5.4 The Rodeo NOAA mission in details

Date	Cycle	Ballast up (ml)	Ballast down (ml)	Pitch	Depth maximum (m)	Details
28/01/2026 21:46:06 UTC	1	500	-500	20°	30	Yo test OK
28/01/2026 21:59:09 UTC	2	500	-500	30°	100	Go to GR01 with mission plan actived
28/01/2026 22:43:40 UTC	4	500	-500	20°	500	
29/01/2026 00:12:51 UTC	5	150	-200	20°	1000	GR01 validated, go to GR02
29/01/2026 02:35:27 UTC	6	200	-300	20°	1000	Alarm 32 because glider was too slow during the descent.
29/01/2026 07:40:35 UTC	7	500	-500	25°	1000	
29/01/2026 16:34:20 UTC	11	500	-500	25°	800	GR02 validated Drift to 800m during 30 min
29/01/2026 19:46:43 UTC	12	200	-470	20°	1000	
29/01/2026 23:34:32 UTC	13	500	-500	30°	1000	Go to GR05
30/01/2026 14:42:43 UTC	21	500	-500	30°	1000	GR05 validated, go to GR03
30/01/2026 20:30:19 UTC	24	500	-500	30°	500	

31/01/2026 00:32:29 UTC	28	500	-500	30°	500	GR03 validated, go to GR04
31/01/2026 01:30:17 UTC	29	500	-500	30°	500	Navigation in waypoint mode
31/01/2026 03:29:07 UTC	31	500	-500	30°	1000	
31/01/2026 07:14:55 UTC	33	80	-380	20°	1000	
31/01/2026 11:19:57 UTC	34	80	-380	20°	1000	GR04 valided, wait the other glider in is point
31/01/2026 23:41:12 UTC	37	80	-380	20°	1000	Go to Bathy (middle of GR03).
01/02/2026 12:14:33 UTC	40	80	-380	20°	1000	Go to GR04
01/02/2026 16:34:04 UTC	41	80	-380	20°	400	Synchronize with the others gliders
01/02/2026 18:33:59 UTC	42	80	-380	20°	1000	Start the transet GR04-GR05
01/02/2026 22:59:44 UTC	43	80	-380	20°	1000	Problem with the paoyload: unable to transfer data from SD to SSD
02/02/2026 11:49:13 UTC	46	500	-500	15°	30	
02/02/2026 12:10:22 UTC	47	500	-500	15°	70	
02/02/2026 13:36:52 UTC	49	80	-380	15°	1000	GR05 validated, go to GR01
03/02/2026 12:28:36 UTC	53	40	-300	15°	1000	

03/02/2026 19:24:10 UTC	54	40	-300	15°	1000	GR01 validated, go to GR05
04/02/2026 02:19:55 UTC	55	80	-300	15°	1000	
04/02/2026 22:34:28 UTC	58	12cm/s	12cm/s	20°	1000	Reboot the glider
05/02/2026 11:09:49 UTC	61	150	-400	20°	1000	
05/02/2026 15:17:15 UTC	62	90	-340	20°	1000	
05/02/2026 19:47:04 UTC	63	90	-340	20°	1000	GR05 validated, go to inter2
06/02/2026 13:56:18 UTC	67	70	-340	20°	1000	Go to GR05
06/02/2026 18:45:11 UTC	68	70	-340	20°	800	GR05 validated, go to GR04. Drift to 800m during 60min.
06/02/2026 18:54:58 UTC	68	-	-	-	-	Reboot the payload
07/02/2026 00:51:36 UTC	69	70	-340	20°	800	Drift to 800m during 20min
07/02/2026 13:01:34 UTC	72	70	-340	20°	800	GR04 validated, go to GR05. Drift to 800m during 20min
08/02/2026 12:04:59 UTC	77	-	-	-	-	Reboot the payload
08/02/2026 14:23:48 UTC	78	50	-310	15°	1000	GR05 validated, go to GR01
09/02/2026 04:31:21 UTC	80	70	-340	20°	1000	

09/02/2026 18:25:37 UTC	83	300	-500	30°	1000	
09/02/2026 22:35:06 UTC	85	300	-500	30°	1000	GR01 validated, go to GR05
10/02/2026 06:10:37 UTC	88	-	-	-	-	Shutdown of the payload
10/02/2026 12:51:45 UTC	92	300	-500	30°	1000	GR05 validated, virtual moring for waiting for recovery
11/02/2026 21:58:24 UTC	159	300	-500	30°	100	Glider recovered

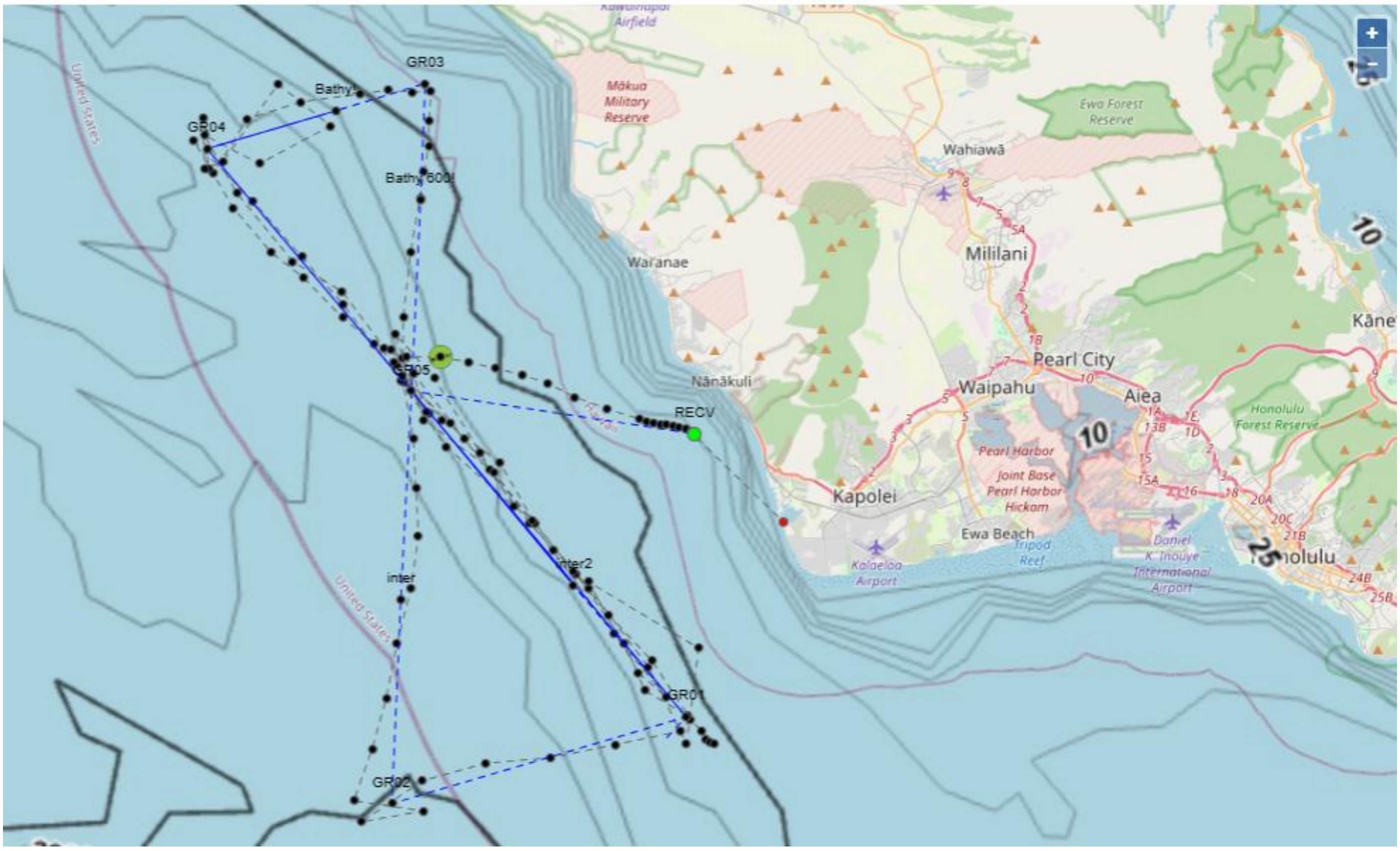

Figure 10: Mission of SEA117 on GLIMPSE

Table 4 : The different phases of the mission

Phase	Date	Type	Details
1	29/01/2026 to 30/01/2026	Trimming	Follow transect GR01 to GR02. Adjust the manual ballast to navigate at 10 cm/s. Perform an easy drift at 800 m to find the drf.ballast value. MV on GR02 until the start of Phase 2.
2	30/01/2026 to 02/02/2026	Fast	Follow transect GR02 to GR03 (caution : 600 m bathy), then GR04. Pass 500 m from the INTER BATHY points, then continue 1000 m toward GR04.
3	02/02/2026 to 05/02/2026	Slow	Follow the GR04 transect towards GR01. Slow down the glider as much as possible, prioritizing acoustic quality during this phase rather than trajectory tracking. Fixed ballast with speed 10 cm/s.
4	05/02/2026 to 07/02/2026	Intermediate	Follow transect GR01 to GR05. Maintain good trajectory tracking and vertical speed ~12 cm/s.
5	06/02/2026 18 :00 UTC to 08/02/2026 18 :00 UTC	Drift	Make a round trip GR05 -> GR04 -> GR05. Maintain good trajectory tracking, vertical speed ~12cm/s, and perform 20-minute drifts at 800m.
6	08/02/2026 06 :00 UTC to 10/02/2026 06 :00 UTC	Mini-rodeo	The goal is to repeat each of the last phases every 12 hours. Navigate along the GR05 -> GR01 -> GR05 transect.
7	10/02/2026 06 :00 UTC	Recovery	Go to the RECV recovery point (be careful of the shallow depth!) and wait there in waypoint mode until recovery. Turn the flasher/radio back on.

5.5 Acoustic raw data

As mentioned in Section 3.2, the raw acoustic data are stored on an SD card and transferred to the external hard drive (SSD) at each surfacing.

From yo 43, the information transmitted in the PLD files indicates that this transfer no longer occurred and that the payload was unable to scan the occupied space on the SSD. Since the SD card has a recording capacity of two days, the payload was rebooted just before the mini-rodeo phase.

After recovery, the SSD did not appear to be corrupted, and it was possible to retrieve the data recorded from 28/01 21:46:06 UTC to 01/02 22:59:44 UTC.

In the end, the raw (.wav) data are available for the following periods:

- From 28/01 21:46:06 UTC to 01/02 22:59:44 UTC
- From 08/02 12:04:59 UTC to 10/02 06:10:37 UTC

It should also be noted that this technical issue did not affect the real-time processing capability of the acoustic data, including ambient noise analysis and marine mammal monitoring.

5.6 Extract of the logbook

The logbook is used by ALSEAMAR pilots to record the various stages of the mission and any changes made to the navigation.

Date	Time	From	Cycle	Message	DOWNLOAD
28/01/2026	21:46:06 UTC	Pierini Camille	1	first test dive at 30m	
28/01/2026	21:58:08 UTC	Pierini Camille	2	payload and navigation are ok	
28/01/2026	21:59:09 UTC	Pierini Camille	2	go to GR01 with mission plan actived, at 100m depth full speed	
28/01/2026	22:43:40 UTC	Pierini Camille	4	I change depth from 100m to 500m and pitch from 30° to 20°	*(modified)*
29/01/2026	00:12:51 UTC	Pierini Camille	5	changed to 1000m with reduced ballast (-200/+150ml)	
29/01/2026	00:27:33 UTC	Pierini Camille	5	GR01 validated, heading toward GR02	
29/01/2026	02:35:27 UTC	Pierini Camille	6	alarm 32 because I was too slow during the descent. I changed to -300 and +200ml	
29/01/2026	09:25:16 UTC	Emmetiere Remi	7	To be discussed with pilot, request spectro: $PLD.AURIS.spectro.request={"start":"20260129-04h53mn30","duration":120}; $PLD.AURIS.spectro.request={"start":"20260129-21h46mn50","duration":60,"fstart":0,"fstop":30000}; $PLD.AURIS.spectro.request={"start":"20260129-07h14mn50","duration":30,"fstart":0,"fstop":30000};	*(modified)*
29/01/2026	09:33:16 UTC	Margirier Felix	7	LEGATO DATA ok, MLD ~100m	
29/01/2026	16:45:24 UTC	Pierini Camille	11	GR02 was reached. Drift underway at 800m	
29/01/2026	19:42:54 UTC	Pierini Camille	12	testing $heading.deadzone=5	
29/01/2026	19:46:03 UTC	Pierini Camille	12	$drf.ballast=-230	
29/01/2026	23:55:44 UTC	Pierini Camille	13	going to GR05 from now	
30/01/2026	13:56:53 UTC	Dufosse Margaux	20	PIC MDU	
30/01/2026	14:22:01 UTC	Dufosse Margaux	20	For the next surfacing, acoustic test change the zb to 30 meters for a short yo	
30/01/2026	14:48:22 UTC	Dufosse Margaux	21	The glider don't get this command zb=30	
30/01/2026	16:24:45 UTC	Heumann Alexandre	21	PIC AHE	
31/01/2026	00:32:29 UTC	Pierini Camille	28	GR03 has been reached	
31/01/2026	00:39:39 UTC	Pierini Camille	28	Change in mission planning in order to wait for the other gliders: after GR04, we return to GR03, then GR04 again	
31/01/2026	06:16:41 UTC	Beguery Laurent	32	Hi everyone, I have change bu=300 in all quick navigation as this glider is a bit light and goin up at 30cm/s is OK for automatic detection but 40cm/s increase the dolfin's detection... Too much flow noise.	
31/01/2026	06:27:21 UTC	Beguery Laurent	32	As we are waiting for the others, I beleive we can start going slow...	
31/01/2026	06:32:45 UTC	Beguery Laurent	32	I have put slow mode with bu=80 and bd=-380	
01/02/2026	03:25:37 UTC	Pierini Camille	37	no need to go to GR03 so I changed mission planning. We are going to "Bathy!" then return to GR04 in slow mode	
01/02/2026	03:27:31 UTC	Pierini Camille	37	When returning to GR04, we will need to set $heading.deadzone=5 for lownoise	
01/02/2026	05:54:08 UTC	Beguery Laurent	38	PErfect, Thanks Camille!!!	
01/02/2026	09:29:07 UTC	Heumann Alexandre	39	The glider should return to GR04 at next surfacing, i add the command in NSD	
01/02/2026	13:39:28 UTC	Heumann Alexandre	40	Start of the transet GR04-GR05 at 6PM GMT. The glider will do a 400 m depth yo at next surfacing to synchronize with the others gliders	
01/02/2026	17:24:23 UTC	Heumann Alexandre	41	Start of the transet GR04-GR05 at next surfacing	
02/02/2026	07:52:31 UTC	Heumann Alexandre	45	Reducing pitch value from 20° to 15° to reduce vertical speed	
02/02/2026	08:53:29 UTC	Desvaux Benjamin	45	PIC BDE	
02/02/2026	08:54:23 UTC	Desvaux Benjamin	45	Halt next surfacing for a payload data check.	
03/02/2026	09:59:28 UTC	Desvaux Benjamin	52	Ballast decreased by 80 mL for descents and 40 mL for ascents on slow profiles.	
03/02/2026	20:07:07 UTC	Emmetiere Remi	54	To discuss with Laurent but slow speed means big science files. With seastate conditions it is hard to complete download in one transfer. Maybe consider to increase speed to 20-25cm/s in order to reduce file size.	
03/02/2026	20:18:35 UTC	Desvaux Benjamin	54	Glider on GR01.	
04/02/2026	00:06:19 UTC	Pierini Camille	54	Selene asks us to turn back toward GR05 with the same settings as on the way there in order to see the difference in navigation (few dives only). So I'm changing the mission plan	
04/02/2026	00:06:47 UTC	Pierini Camille	54	(we are ahead of the other gliders)	
04/02/2026	10:20:49 UTC	Haftman Alexandre	56	PIC AHA	
04/02/2026	22:19:55 UTC	Haftman Alexandre	58	reboot of the payload	
05/02/2026	10:22:18 UTC	Haftman Alexandre	60	bu of the navigation profile modified for next dive in order to reduce the speed ascent to 12 cm/s. (from automatic 210 mL (1012) to fix150 mL). rVal set to 0 to stay on virtual mooring on Gr05 until 07/02.	*(modified)*
05/02/2026	16:01:15 UTC	Haftman Alexandre	62	Due to an advance of 1 day, I add a step to inter2 and a step back to GR05. The goal is to be back to GR05 before 07/02.	*(modified)*
06/02/2026	11:09:20 UTC	Renou Melvyne	66	PIC MRE	

06/02/2026	18:54:58 UTC	Renou Melvyne	68	Reboot of the payload
06/02/2026	18:59:03 UTC	Renou Melvyne	68	Start the phase 5 with drift mode in zb=800 during 1 hour going to GR04
07/02/2026	00:31:10 UTC	Renou Melvyne	69	drf.ballast=-190
07/02/2026	00:32:03 UTC	Renou Melvyne	69	changing the duration of the drift at 20min
07/02/2026	12:51:19 UTC	Renou Melvyne	72	drf.ballast=-210
07/02/2026	12:51:48 UTC	Renou Melvyne	72	GR04 reached go back on GR05
08/02/2026	12:04:59 UTC	Renou Melvyne	77	reboot of the payload this morning
08/02/2026	15:04:28 UTC	Renou Melvyne	78	GR05 reached in drift mode after this drift the glider can go Slow navigation mode to GR01
08/02/2026	22:38:18 UTC	Renou Melvyne	79	Go for a dive in Slow mode navigation
08/02/2026	22:40:20 UTC	Renou Melvyne	79	next dive in Intermediate navigation mode loaded in Mission plan
09/02/2026	08:30:40 UTC	Pierini Camille	80	Last payload reboot 08/02/2026 - 06:04 UTC
09/02/2026	17:06:55 UTC	Margirier Felix	82	Mode FAST for next surfacing
10/02/2026	06:10:37 UTC	Margirier Felix	88	$pld1.enable=0
10/02/2026	11:08:35 UTC	Margirier Felix	91	virtual mooring on GR05 waiting for recovery
10/02/2026	20:49:44 UTC	Beguery Laurent	96	recovery for tomorrow; will do 100m

CONCLUSION

The glider was ballasted a bit light for the region. With a better ballasting the max speed of the glider could have been slightly higher. This did not affect the mission.

The SeaExplorer slow mode was set to avoid flow noise in the recording.

The problem on the hard drive left us with only 6 days of recording instead of the 14 expected. During those missing days, the detections algorithms and background noise were still calculated.

Apart from the point above, the preparation and execution of the mission was conducted successfully.

APPENDIX

Appendix 1: SeaExplorer main features

SPECIFICATIONS

Body size (D x L)	0.25 m x 2 m + 0.7 m foldable antenna
Wingspan	56.5 cm. Wingless design
Weight	59 kg in air
Ballast volume	1 L (± 500ml)
Speed	Up to 1 knot horizontal
Payload	9 L / 8 kg in two sections (wet/dry)
Architecture	2 independant low-power CPUs (Linux) for payload & navigation
Embedded soft-ware	Payload: Opensource C++ / Linux Navigation: Proprietary
Depth rating	700 m (850 m survival)
Pitch in navigation	± 15 to 40° (± 20° typical)
Turn radius	20 m (allows virtual mooring)
Battery	Rechargeable Li-ion / Primary cell optional
Battery endurance	Up to 2 months (Depending on sensors configuration)
Recharging time	20 hours
Communications	Triple antenna with strobe light (by default) GPS / Satellite (Iridium) / Radio
Local Radio range	1 km @ 902 to 928 MHz
Data format	Compressed CSV (native)
Data downloading	Ethernet cable through external connector (no vehicle opening)
Safety	Autonomous Drop-weight Option: Locator Pinger (ULB) and/or Argos
Sensors	4 "puck type" ports available
Available Sensors	Conductivity Temperature Pressure (Sea-Bird GPCTD) Dissolved Oxygen (Sea-Bird SBE 43F) Chlorophyll (Wetlabs ECO Puck) CDOM (Wetlabs ECO Puck) Turbidity (Wetlabs ECO Puck) Hydrocarbons (MinFluo-UVc sensor exclusivity) Methane (Franatech METS) Sewage & Pesticides (MiniFluo-UVc sensor exclusivity) Acoustic Recorder ADCP Nitrates Altimeter Others upon request

BRIDGES

ALSEAMAR has been funded by EU to develop ultra-deep gliders for 2,400m and 5,000m depth. Find out more about the BRIDGES project on www.bridges-h2020.eu

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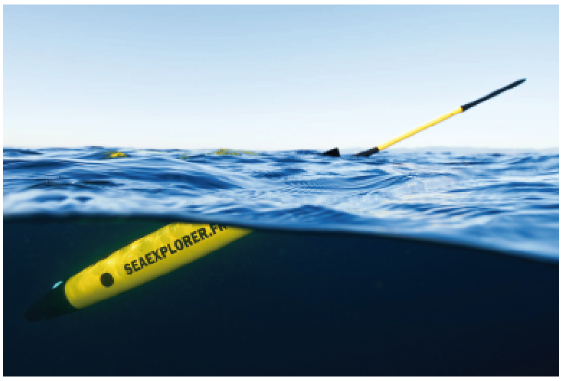
ALSEAMAR ALCEN

SEA EXPLORER Underwater Glider

Low-logistics & Multi-mission glider



Monitoring
Measuring
Detecting
Recording

APPLICATION FIELDS

Oceanography & Science
Environmental Research & Monitoring

Oil & Gas
Exploration & Environmental Baseline Studies

Defense & Security
Acoustic Monitoring & Patrolling

innovation & **services** at **sea**

Drive system

The SeaExplorer underwater glider is driven by three internal actuators for propulsion, pitch and roll. These actuators enable the glider to move independently both horizontally and vertically.

Propulsion

The SeaExplorer propulsion is based on the principle that materials with a higher density than water will sink, while those with a lower density are buoyant. Materials with the same density remain stable in water. The SeaExplorer changes its density via an external oil bladder. When the oil bladder deflates, the glider dives and glides using the lift provided by the fins. When the oil bladder inflates, the glider climbs to the surface and glides using the lift provided by the fins. This results in a saw tooth pattern. The actuator for bladder inflation and deflation for propulsion is the ballast.

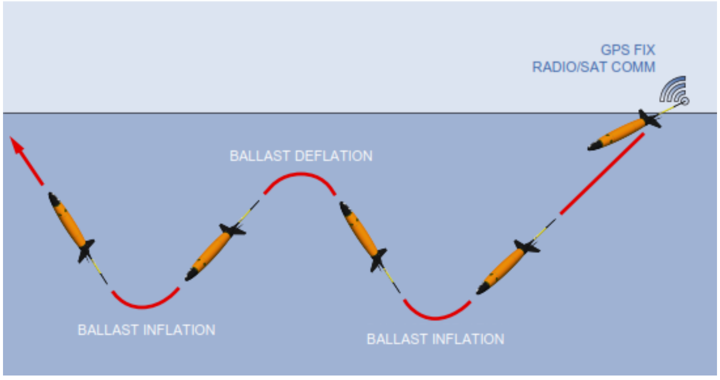
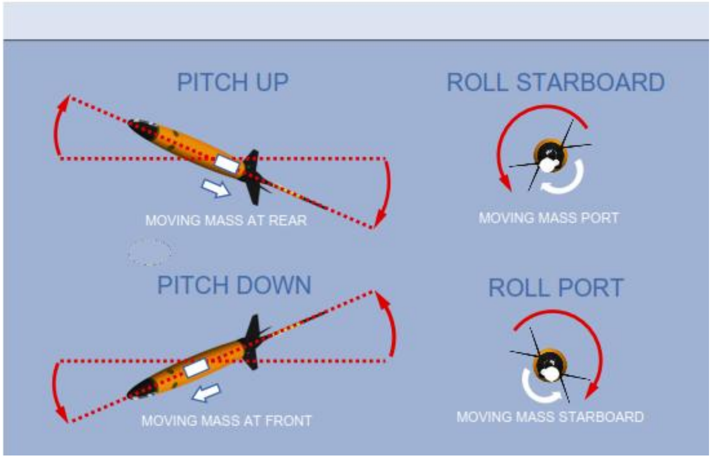


Figure 2: Glider propulsion mode

Attitude

The SeaExplorer is also driven by two other internal actuators for the attitude. The first actuator displaces the moving mass forward and backward to create up/down pitch and the second rotates the moving mass to port or starboard for roll control (Figure 3). The roll control enables the glider to follow or change its heading.



Glider attitude mode

Communication

Upon surfacing, the GPS position, navigation data together with sensor data are transmitted via the Iridium satellite to the onshore Supervising and Piloting Station (SPS). In return, navigation set-points can be sent back to the SeaExplorer for any necessary mission modifications. For closer communications (with ship or in lab), the SeaExplorer is fitted with a Radio Frequency (RF) system. As optional devices, the user can install an Argos beacon and/or an underwater pinger for other means of localization.

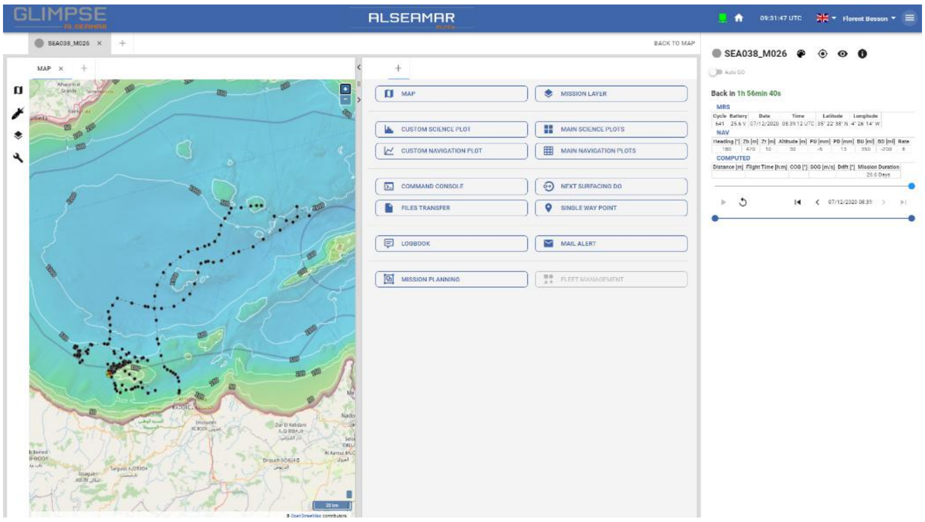
Piloting

GLIMPSE : Glider Mission Piloting System for Explorers

GLIMPSE is a secured online tool for glider piloting and supervision. The platform is restricted to SeaExplorer user's and customers. It stands for GLIder Mission Piloting System for Explorers.

This platform is accessible via a webpage, internet will be required.

URL address is: https://www.glimpse-alseamar.com



GLIMPSE web interface

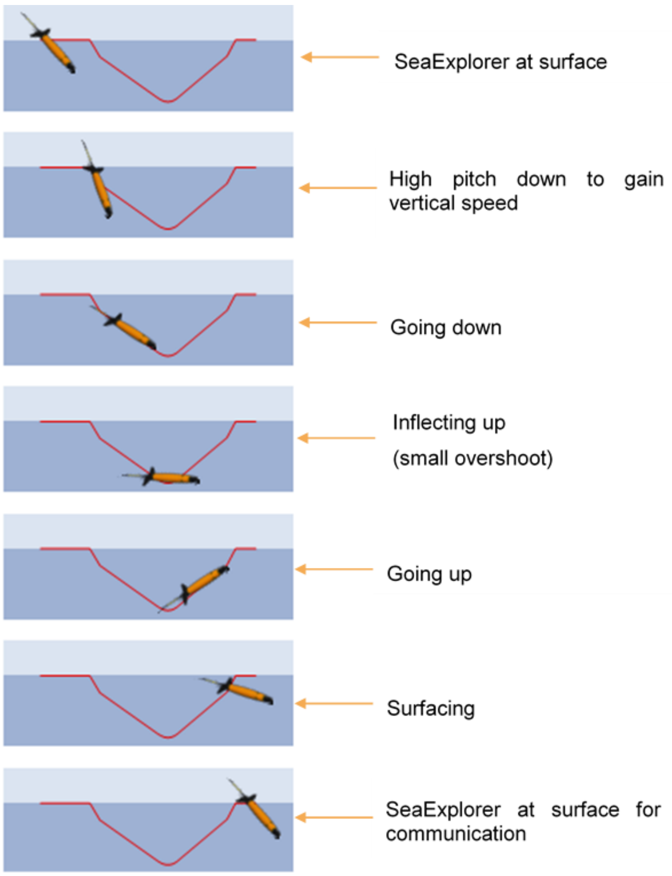
Power supply

A Li-ion battery pack powers the actuators, electronics, navigation sensors, communication devices and the payload section. Three types of battery are available, rechargeable 1st (deprecated) and 2d generation, and non-rechargeable battery pack. To charge the batteries, the operator must plug the charger cable directly into the vehicle in the wet aft section without opening the vehicle. The battery pack is situated in the central section.

Standard Navigation sequence

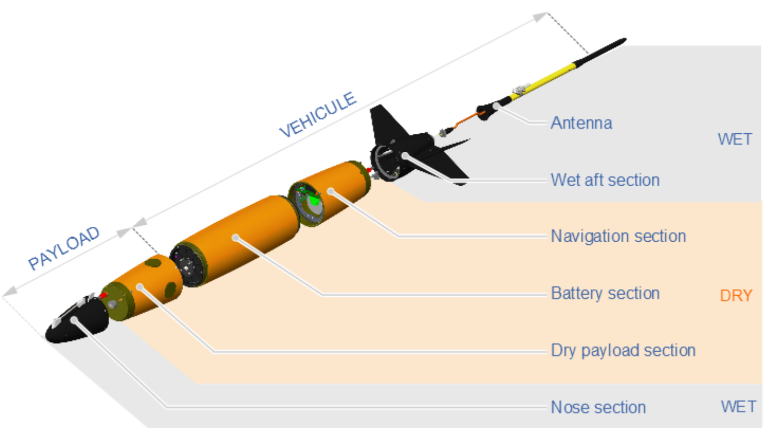
A standard navigation cycle is called a “yo”. When the glider is working at sea, the standard navigation cycle is the following: the glider is at the surface, receiving time and position through GPS and communicating at the same time with the pilot through Iridium. Once authorized by the pilot, the glider deflates the bladder and runs the moving mass forward to pitch down and dive. During the first meters, the glider will dive with a high angle to gain vertical speed before adjusting its pitch to glide. Once the required depth is reached, the glider will inflate its ballast and run the moving mass backward for pitching up, to overshoot, climb and reach the surface. Depending on the programming, the glider can either dive for another yo or return to the surface for communication.

Navigation sequence



SeaExplorer breakdown

The SeaExplorer comprises the payload part (Front) and the vehicle part (Back). The vehicle part is terminated by two x-shaped fins and extended by the antenna mast.



SeaExplorer Breakdown

The body includes 6 main sections:

The nose section:
Dedicated to the integration of flooded sensors. It can be fitted with GPCTD pumped sensors from Seabird (Conductivity Temperature, Depth), optical windows, inlets/outlets, acoustic devices, etc.

The dry payload section:
Dedicated to the integration of dry sensors including four PUCK ports, payload electronics processing (dedicated open-source CPU), altimeter, water inlet sensor and balancing weights.

The battery section:
Includes the battery pack providing power, and the moving mass mechanism to set the vehicle attitude (the battery pack is used as a moving mass).

The navigation section:
This section includes the ballast engine and the navigation system (Navigation board and sensors), water inlet sensor.

The wet aft section:
Composed of the oil bladder and the release system in addition to connectors for charging, powering up and onshore data transfer.

The antenna.

The body is composed of one dry compartment (pressure resistant) and two wet sections located fore and aft.

For handling and transport, the antenna can be folded along the body of the glider and the glider is fitted with two built-in rings (located back and front) as lifting points.

Appendix 2 : Sensor datasheet and calibration sheet

LEGATO CTD datasheet

Summary:

Sensor name	Manufacturer	Parameters
LEGATO	RBR	Conductivity, temperature, density
CODA	RBR	Dissolved oxygen, temperature, corrected phase

The RBR LEGATO is a CTD that does not require a pump. The sampling period of this sensor can be configured in 1,2 or 16 Hz. Some version of the LEGATO has a CODA sensor connected to measuring dissolved oxygen. Even with a CODA sensor, the interface between payload and sensors is done by Legato.

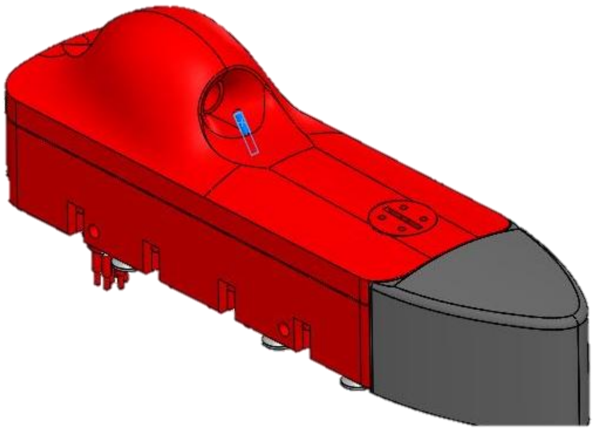


Figure 1 RBR CODA sensor

Figure 2 RBR LEGATO sensor

Configuration:

Items are prefixed with “cfg.”.

Parameter	Recommended value	Description
WarmUpPeriod	[Min ; Max] = [4s ; 10s] Default = 4s	Time in seconds between start-up and communication.
Phaseswitch	1	Defines the ability of the sensor to be turned on/off according to flight phase. Value can be 0 or 1: • 0 means “sensor always on” • 1 means “sensor is allowed to be switch off”. Use 0 if the sensor technology forbids to turn off the sensor at sea.
Periodswitch	0	Defines the ability of the sensor to be turned on/off between each measurement. Value can be 0 or 1: • 0 means “sensor always on” • 1 means “sensor is allowed to be switch off”. If period switch=1, the sensor will switch OFF only if WarmUpPeriod < acq.X.period

Parameter	Recommended value	Description
codaInstalled	1	Indicate to payload software if coda sensor is connected to Legato. • 1 means coda is installed • 0 means no coda is installed
log_legato	1	This parameter is useful if the Legato is a fast version (16Hz). • 1 means that the software will create a "legato.raw" log file as fast as possible • 0 means that the software will only create pld.raw and pld.sub log files.
acq.x.period	0	Period (in seconds) with which the payload will read sensor data. 0 means as fast as possible (recommended for fast sensor version)

Data output:

Parameter	Format	Units	Description
LEGATO_CONDUCTIVITY	Float	mS/cm	Conductivity
LEGATO_TEMPERATURE	Float	°C	Temperature
LEGATO_PRESSURE	Float	dbar	Pressure
LEGATO_SALINITY	Float	ups	Salinity

If coda is installed, 3 more fields will be log in files:

Parameter	Format	Units	Description
LEGATO_CODA_DO	Float	µmol/L	Dissolved oxygan
LEGATO_CODA_TEMPERATURE	Float	°C	Temperature
LEGATO_CODA_CORR_PHASE	Float		Corrected phase

Connection:

LEGATO sensor has to be connected to a RS232 serial link.

- Baud rate : 115200
- Data bits : 8
- Parity : none
- Stop bits : 1
- Flow control : none

LEGATO CTD calibration sheet

SN 230630

RBR Conductivity Calibration Certificate

RBRlegato³ C.T.D, Alseamar, wet bay (1000dbar) s/n: 230630
References: Autosal8400B#66289, MS-315#15506, SSW P166, RC#002

Reference Resistance (ohm)	Reference Conductivity (mS/cm)	Voltage Ratio, V	Measured Conductivity (mS/cm)	Calibration Error (mS/cm)
open	0.0000	-0.000219	0.0002	0.0002
694.040	7.1715	0.038845	7.1713	-0.0002
331.930	14.9951	0.081463	14.9949	-0.0002
150.020	33.1776	0.180517	33.1784	0.0008
100.015	49.7656	0.270872	49.7653	-0.0003
75.023	66.3438	0.361179	66.3432	-0.0006
55.520	89.6489	0.488137	89.6493	0.0004

Bath	Voltage Ratio	Temperature (ITS-90)	Salinity (PSS-78)	Conductivity (mS/cm)
T15S35	0.2336489	15.02571	34.9902	42.9321
T25S35	0.2923994	25.62744	34.9884	53.7116

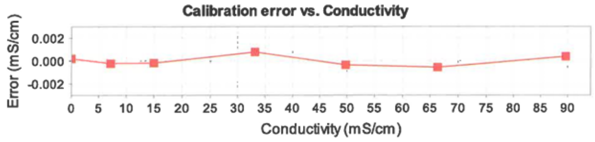
Cell Constant @T15S35 = 4.97731 1/cm

Coefficients

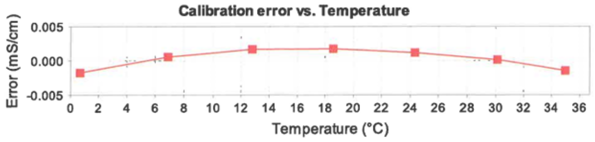
	C0:	40.384352E-3
	C1:	183.57335
(K)	C2:	1.0
	X0:	1.0293382E-3
	X1:	-9.540584E-6
	X2:	449.7304E-9
	X3:	-173.3105E-12
	X4:	53.309E-15
	X5:	15.025705
	X6:	10

$$C_c = \frac{C_0 + C_1 * C_2 * V - X_0 * (T - X_5)}{1 + X_1 * (T - X_5) + X_2 * (P - X_6) + X_3 * (P - X_6)^2 + X_4 * (P - X_6)^3}$$

Calibration error vs. Conductivity



Calibration error vs. Temperature



RBR Temperature Calibration Certificate

Logger ID: RBRlegato³ Serial No: 230630 Channel No: 2

Reference Temperature, ITS-90	Voltage ratio, V	Measured Temperature, ITS-90	Calibration error
-4.97075	0.7130195	-4.97071	0.00004
0.67461	0.6483895	0.67450	-0.00011
6.90098	0.5732431	6.90103	0.00005
12.80883	0.5013629	12.80888	0.00005
18.57749	0.4335115	18.57751	0.00002
24.32259	0.3704649	24.32251	-0.00008
30.11704	0.3129590	30.11703	0.00000
35.00254	0.2697982	35.00256	0.00002

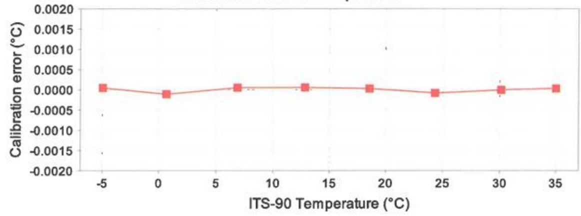
Coefficients

C0:	3.4956217E-3
C1:	-253.94728E-6
C2:	2.4712892E-6
C3:	-85.088246E-9

$$T_m = \ln\left(\left(\frac{1}{V}\right) - 1\right)$$

$$T_c = \frac{1}{(C_0 + C_1 T_m + C_2 T_m^2 + C_3 T_m^3)} - 273.15$$

Calibration error vs. Temperature



RBR Pressure Calibration Certificate

RBRlegato³ C.T.D, Alseamar, wet bay (1000dbar) s/n: 230630
Instrument rating: 1000 s/n: P217465
Nominal accuracy: 0.05%FS (0.5 dbar)
Reference instrument: Mensor CPC6050 s/n: 41000CAM

Applied pressure, P_{app} (dbar)	Voltage ratio, V	Measured pressure, P_c (dbar)	Calibration error (dbar)
10.232	0.0195703	10.239	0.007
110.269	0.0626006	110.277	0.008
210.269	0.1055658	210.271	0.002
310.267	0.1484891	310.263	-0.004
410.268	0.1913848	410.278	0.010
510.268	0.2342374	510.266	-0.002
610.267	0.2770666	610.263	-0.004
710.267	0.3198775	710.270	0.003
810.267	0.3626683	810.272	0.005
910.266	0.4054479	910.277	0.011
1010.280	0.4482201	1010.285	0.005

Coefficients

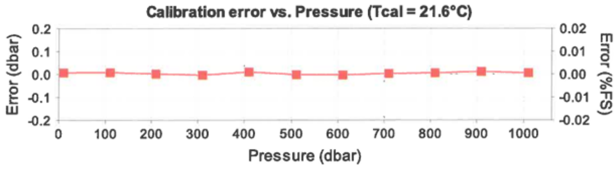
C0:	-35.22914
C1:	2.322284E3
C2:	33.24219
C3:	-22.947985
X0:	9.9623
X1:	57.464458E-3
X2:	25.804024E-6
X3:	-53.00522E-9
X4:	-140.91836E-6
X5:	21.599476

Head (mm) = 270

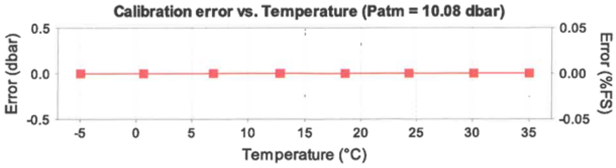
$$P_c = X_0 + \frac{P_m - X_0 - X_1(T - X_5) - X_2(T - X_5)^2 - X_3(T - X_5)^3}{1 + X_4(T - X_5)}$$

$$P_m = C_0 + C_1 V + C_2 V^2 + C_3 V^3$$

Calibration error vs. Pressure (Tcal = 21.6°C)



Calibration error vs. Temperature (Patm = 10.08 dbar)

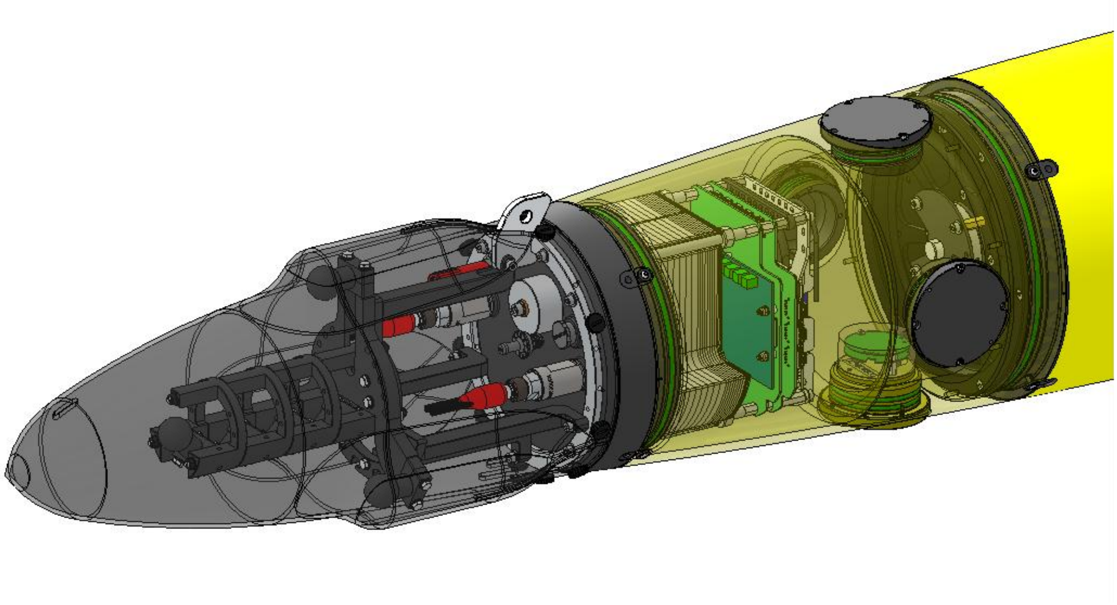


AURIS acoustic antenna datasheet

The acoustic payload developed by ALSEAMAR includes :

- The AURIS platform with a Jetson Nano computer
- An ADC and its distribution board
- Pre-amplifiers
- 4 hydrophones

A complete manual is dedicated to this payload, please refer to it for more details: SEF00005038-AURIS Payload - User manual revG



GLIDER PAYLOAD WITH AURIS SENSOR

Appendix 3 : SEA117 ballasting sheet

Buoyancy and Trim Spreadsheet

Product:	**SeaExplorer** N°	SEA117	S/N	
Writer:	CPI		Version:	E-28/08/2019
Object :	Sea Explorer buoyancy and Trim setting			
Mission :	NOAA			
Date:	08/10/2025	Ref fichier :	Y:\ENGINS_NON_HABITES\PRODUIT\SEA_EXPLORER\8-Suivi materiel missions ventes_Missions Service ALSEAMAR\2025-10-08_SEA117_NOAA\[Glider-NDC-002-SEA117 1023 Glider buoyancy calculation-RevF.xls]Buoyancy processing	

Commercial In Confidence

Reminders:

1 inox washer (g) =	29									
water density	1,028									
Inox density	8,01	29	g in the air is	25,28	g in water	1	g to be added in water is	1,15	g in air	Number of inox to be added	0,04
Foam density =	0,38	1	g in the air is	-1,71	g in water	1	g to be removed in water is	-0,59	g in air	grams of foam to be added	-0,59

Configuration of the payload

Nose sensor		
Top sensor		
Bottom sensor		
starboard sensor		
Port sensor		
Payload weight - number of 250 g plates		optional
Payload weight - number of 100 g plates		optional
Payload weight - number of 50 g plates		optional
Weight of the Payload		optional

For ALSEAMAR team
Do not forget to complete the file SEA06_ConfigCU_lestage : Y:\ENGINS NON HABITES\PRODUIT\SeaExplorer\8-Suivi materiel missions ventes_Suivi SAT

Configuration of the vehicle

Weight of the ballast section		optional (with release and spring, not ULB not ARGOS)
Weight of the battery section		optional (with release and spring)
Total weight of the glider (kg)	61,088	(with payload, release weight, spring - without foam/washers)
Argos beacon	YES	
ULB	NO	Position: Nose

Ballasting

Ballast water density=	1000	kg/m3	(ρ)
Ballast water temperature	20	°C	
Target Sea mass density=	1023	kg/m3	(ρsea)
Target Sea temperature	15	°C	

	(Pw)	(Fw)	(P1)	(P2)	(Pwater)	V-(Pair+Pw+Fw-Pwater)/ρ	Psea-Pair+Pw+Fw-V*ρsea			
	Number of washers added (29g in air)	Foam weight in the air added (g)	Weight in fresh water - Front (kg)	Weight in fresh water - Rear (kg)	Total weight in fresh water (kg)	Volume of the glider (l)	Weight of the glider at sea (g)	Battery position (mm) Target is ~35-40 mm Min/Max is 25/53	Glider angle measured when angular set to +45°	Glider angle measured when angular set to -45°
Ballasting measurement			0,67	0,7	1,37	59,70	15	38		
	Nose	Nose					Glider will sink			
							Glider is well ballasted			

Key: Input Output

Appendix 4 : SEA117 mission preparation sheet

ALSEAMAR ALCEN

OBJECTIVE : Mission NOAA

Author : RSA
Date : 24/10/2025

GLIDER CONFIGURATION			
Glider N°	SEA117	**Glider Version**	3.8.2-r

TESTER MODE AFTER GLIDER CLOSING	
Release state (armed)	Armed ☒ Depth sensor used : nav
Drop test with tool	☒
Release configuration	Software version: v1.2.0 Release pressure sensor: 1 Nav pressure sensor: 1 Maximum depth: 1050 Watchdog timeout: 900s Time / depth stability: 1800s/10m Drop channel: 1
Compass	☒
Vacuum (76 000 / 80 000 Pa)	☒
Flasher (0/1)	☒
Ballast (+/- 500 ml)	☒
Linear (0-100 mm)	☒
Angular (+/- 80°)	☒
Depth (+/-10m)	☒
Specific parameters : Dry air used = no	

SEA.MSN PARAMETERS (for next simulation)	
mission.num N+1 and mission mode 3	☒
Specific parameters :	

SEA.CFG PARAMETERS	
altimeter.type	1 (Alti)
battery.coef (X2.2)	;
battery.zero (X2.2)	;
pld1.install	3 (Auris)
pressure.range	105 (1050)
security.batteries.low	23 (18650)
security.depth.limit	1025
Specific parameters :	

RunTester – thrusters	
o (allumer l'option)	☐
b (allumer le thruster babord, tourne 3secs)	☐
t (allumer le thruster tribord, tourne 3secs))	☐
B (set a speed at 10%) – turns clockwie If wrong way, change *thrurster.babord.invert*	☐ *thrurster.babord.invert=*
T (set a speed at 10%) - turns clockwie If wrong way, change *thrurster.tribord.invert*	☐ *thrurster.tribord.invert=*
s (stop the option)	☐

SEAPAYLOAD.CFG FILE			
Payload number	Pld666	**Payload version**	2.1.0.r / 1.0.13.r
AD2CP mission number +1	☐		
SLOT	SENSOR	S/N Vendor	Date of last calibration
[4]			
[5]	Legato	230630	Juillet 2023
[6]			
[7]			
[8]			
[9]			
Specific parameters : LEGATO fast oui ou non, si oui vérifier le bruit durant la mission.			

ARGOS		PINGER	
S/N	24U2928	S/N	69605
PTT	266174	Ping ?	☒
Energy level (>3V)	☒		
Mode auto start	☒		
Suivi Argos -Pinger.xlsx	☒		

Ballasting		Compass	
Buoyancy calculation	☒	Compass calibrated	☒
Density	1028	AD2CP calibrated	☐
Thruster on/off	no		

SIMULATION & PREPARATION			
Previous mission saved (CCN/CCU)	☒	Simulation number	21
Previous mission deleted on glider	☒		
SIMULATION			
GPS position	☒	Sci/nav data check	☒
5 YO's completed to 30m	☒	Data Saved on SST	☒
$heading=1010	☒	Data Deleted	☒
Data transfer on GLIMPSE	☒		
1 yo thruster : $drf.enable=1 ; $drf.timeout=10 ; $drf.speed 50		Check the thrusters duri,g the drift ☐	

Actuators in neutral position	☐
mission.num=+1 / mission.mode=0	☒
Sci/Nav configs saved on SST	☒

VISUAL CHECK	
Locking pins	☐
Antenna	☐
Cover front	☒
Cover rear	☐

UPDATE SEA06_recapitulatif_tests_en_mer.xls ***DO NOT FORGET THE SN OF SENSORS***	☒

AT SEA - GLIDER START-UP

NMEA SENTENCE	
$SEADST : Vacuum between 76 000 / 80 000 Pa	☐
$SEANAV : Energy level over 28 V	☐
NO $SEAMUL appears	☐

ARGOS	
Start mode (10 flashs)	☐

COMMANDS	
?release.status !release.status=0;	☐

LAST CHECK	
Plugs & Locking sleeves	☐
Rope & Buoy	☐
Sensor Caps	☐

HELP

Appendix 5 : SEA117 calibration sheet

ALSEAMAR	MUM	SEA-MUM-003- Compass calibration report - VA.doc
		Revision : C
	CR Compass Calibration	Date : 08/09/2025
		Page : 1/1

Job N°:		
P/N :		
S/N:	SEA117	
Date :	08/10/2025	
Author :		
Procedure reference		**SEA06-50-MUM-UsermanualSeaExplorer-revB**
Equipment ID used :		
Laptop and ethernet calble		

1	**Cardinal points before calibration**	**C***	**NC***	**NT***
1.1	North: ____4_______	☐	☒	☐
1.3	East: ____90_______	☐	☒	☐
1.5	South: ____184_______	☐	☒	☐
1.7	West: ____280_______	☐	☒	☐
	If all values are within +/-3°, the calibration is ok and do not need to be redone.			
2	**Cardinal points after calibration**	**C***	**NC***	**NT***
2.1	North: ____0_______	☒	☐	☐
2.3	East: ____91_______	☒	☐	☐
2.5	South: ____180_______	☒	☐	☐
2.7	West: ____269_______	☒	☐	☐

Note the Glider Temperature during calibration:___11___

For PRIME PRO: note the magnetic calibration score:______ (<1)